VTTSI-Chat: Agentic GeoAI for Real-Time Intersection Safety Monitoring with Natural Language Explanation [Demo]

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Abstract

We present **VTTSI-Chat**, a cloud-native agentic GeoAI system that pairs the Virginia Tech Transportation Safety Index (VTTSI)—a hybrid Real-Time Safety Index (RT–SI) and CRITIC-weighted MCDM pipeline—with an integrated Vite React dashboard and a **SafetyChat** conversational module powered by a tool-augmented GPT-4o. VTTSI fuses seven years of VDOT crash records via Empirical Bayes stabilization with live J2735 connected-vehicle telemetry stored in a PostGIS spatial database, producing interpretable 0–100 safety scores at 15-minute resolution across 18 geospatially registered intersections in Falls Church, VA (three currently streaming live telemetry; the remaining fifteen are wired into the pipeline and awaiting sensor activation). The production dashboard combines live KPI cards, an OpenStreetMap risk map, trend analysis, sensitivity analysis, a database explorer, and the chat interface in one operational workspace. SafetyChat exposes six typed API tools to the LLM, including a guarded read-only SQL tool, grounding all natural language responses in live geospatial data and enabling TMC operators, planners, and emergency responders to query causal safety narratives through ordinary conversation. A live deployment at https://safety-index-frontend-180117512369.europe-west1.run.app/ demonstrates four end-to-end use cases and is available for interactive exploration at the conference.

CCS Concepts

• **Computing methodologies** → **Natural language processing**; **Intelligent agents**; • **Information systems** → *Decision support systems*; • **Human-centered computing** → *Interactive systems and tools.*

Keywords

agentic GeoAI, traffic safety, spatio-temporal safety index, natural language interface, MCDM, Empirical Bayes, connected vehicles, PostGIS, Vite dashboard, tool-augmented LLM

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1 Introduction

Connected-vehicle infrastructure increasingly produces real-time intersection safety signals, but traffic management center (TMC) operators still need explanations, not only numeric scores. Prior LLM+traffic systems focus on signal control [5] or AV planning [6, 13]; they do not explain live infrastructure safety indices. **VTTSI-Chat** fills this gap with an agentic GeoAI interface that pairs the Virginia Tech Transportation Safety Index (VTTSI) with **SafetyChat**, a tool-augmented GPT-4o module that calls typed PostGIS-backed API endpoints and returns grounded spatial narratives. The live system monitors 18 geospatially registered intersections in Falls Church, Virginia, exposes a Vite React dashboard, and supports interactive queries for operators, planners, responders, and public stakeholders.

2 VTTSI System Overview

The deployment uses a Vite React frontend, FastAPI backend, and Cloud SQL PostgreSQL/PostGIS, all served from Google Cloud Run. Six VTTI Trino streams—`bsm`, `psm`, `vehicle-count`, `vru-count`, `speed-distribution`, and `safety-event`—are synchronized into 15-minute bins. VDOT crash records from 2017–2024 are spatially joined to intersection geofences through PostGIS `ST_Within`; overlaps are resolved by nearest centroid. The backend caches frequent read endpoints, computes two complementary indices, and serves both dashboard panels and SafetyChat tools. The complete RT–SI/MCDM derivation, including uplift-feature definitions, CRITIC weights, SAW/EDAS/CODAS normalization, and parameter validation, is documented in the companion VTTSI paper [3]; this demo paper restates the operational equations needed to understand the live system.

2.1 Real-Time Safety Index (RT–SI)

RT–SI anchors intersection risk to a severity-weighted Empirical Bayes (EB) baseline derived from 2017–2024 VDOT crash records. The historical crash rate is:

$$r_i = \frac{\sum_s \ w_s \, C_{i,s}}{E_i}, \quad w_{\text{Fatal}} = 10, \ w_{\text{Inj}} = 3, \ w_{\text{PDO}} = 1, \tag{1}$$

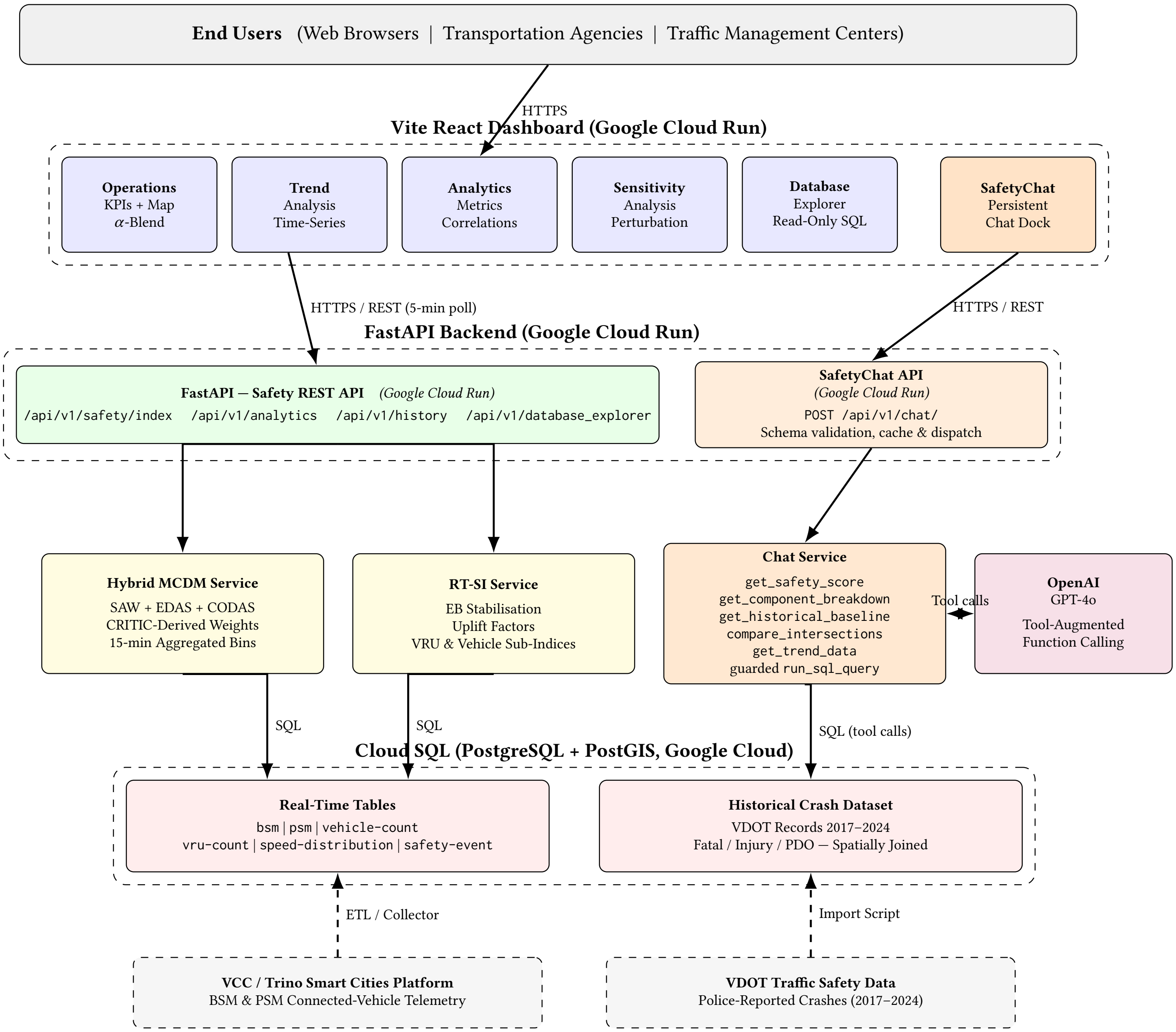


Figure 1: VTTSI cloud-native system architecture. The orange pathway (SafetyChat API, Chat Service, OpenAI GPT-4o) is the new contribution of this demo paper; the blue/green/yellow left pathway shows the existing safety-index pipeline.

where $C_{i,s}$ are crash counts by severity and E_i is exposure. The EB stabilized rate is:

$$\hat{r}^{\mathrm{EB}}_{i,t}(\lambda) = \frac{Y_{i,t} + \lambda\, r_0}{1+\lambda}, \quad \lambda^{\star} = 100{,}000, \tag{2}$$

where r_0 is the global mean and $\lambda^{\star}$ minimizes 2025 hold-out Poisson log-loss. Bounded real-time uplift factors adjust this baseline for speed deficit, speed variance, and VRU–vehicle conflict exposure:

$$U_{i,t} = 1 + \beta_1 F^{\mathrm{spd}}_{i,t} + \beta_2 F^{\mathrm{var}}_{i,t} + \beta_3 F^{\mathrm{conf}}_{i,t}, \tag{3}$$

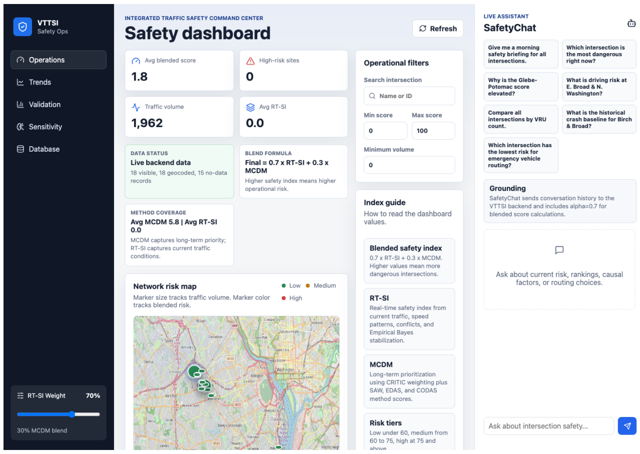
with $(\beta_1, \beta_2, \beta_3) = (0.3, 0.3, 0.4)$; VRU and vehicle sub-indices are blended by $(\omega_{\mathrm{vru}}, \omega_{\mathrm{veh}}) = (0.6, 0.4)$ and capped to $\mathrm{SI}^{\mathrm{RT}} \in [0, 100]$.

2.2 MCDM Safety Index

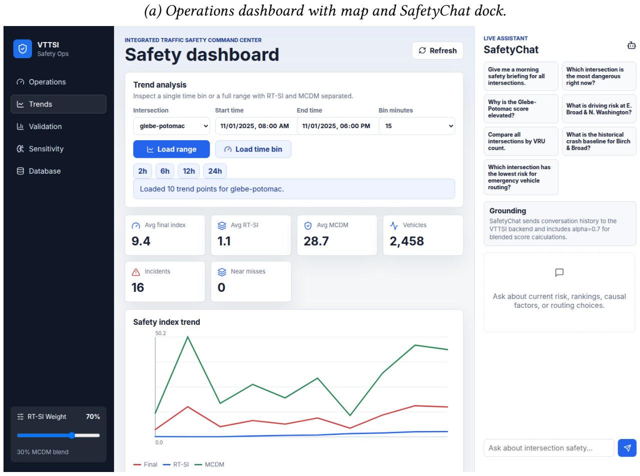
A CRITIC-weighted MCDM pipeline [1] scores each intersection–time bin over vehicle count, VRU count, average speed, speed variance, and incident count. SAW, EDAS, and CODAS are scaled to $[0, 100]$ and combined by a second CRITIC pass into $\mathrm{SI}^{\mathrm{MCDM}}$. Across 18 intersections, method correlations are $r \approx 0.78$–0.99; RT–SI changes by only ≈ 0.09 mean absolute points under $\pm 25\%$ joint parameter perturbation [9].

2.3 Blended Final Index

$$\mathrm{SI}^{\mathrm{Final}}_{i,t} = \alpha\, \mathrm{SI}^{\mathrm{RT}}_{i,t} + (1-\alpha)\, \mathrm{SI}^{\mathrm{MCDM}}_{i,t}, \quad \alpha = 0.7 \text{ (default)}. \tag{4}$$



(a) Operations dashboard with map and SafetyChat dock.



(b) Trend analysis panel in the same workspace.

Figure 2: Live VTTSI-Chat demonstration after the Vite migration: (a) integrated operations dashboard with map and chat combined; (b) trend-analysis workflow retained inside the same interface.

Scores map to risk levels: 0–40 = low, 41–70 = moderate, 71–100 = high.

2.4 Geospatial Foundation and Dashboard

Each intersection has a centroid and bounding-box geofence ($\delta \approx 0.001^\circ$, $\approx$111 m). The Vite dashboard polls `/api/v1/safety/index`, renders an OpenStreetMap risk map, colors markers by risk tier, scales markers by traffic volume, and exposes an α slider, trend analysis, sensitivity analysis, database exploration, and a persistent SafetyChat dock (Figure 2).

3 SafetyChat: LLM-Powered Explanation Module

VTTSI answers *what* the score is; SafetyChat answers *why* and *what to do*. GPT-4o uses typed function calls [8] in an agentic loop of up to six tool calls. The tools expose RT–SI/MCDM/blended scores, component breakdowns, historical EB baselines, intersection comparisons, 7–30 day trends, and a guarded read-only SQL interface over PostGIS. The SQL tool rejects write/DDL keywords, SQL comments, and multiple statements; backend paths use SQLAlchemy, parameter binding, allowlisted identifiers, or literal escaping as appropriate.

Table 1: Summary of Demonstration Use Cases

Use Case	Interaction and grounded response
Morning briefing	Ranks all intersections, drills into the top sites, and summarizes peak-risk windows and dominant factors.
Causal explanation	Replays a historical bin and traces a high score to vehicle volume, incidents, speed variance, or VRU exposure.
Responder routing	Compares two candidate crossings and recommends the lower-risk option using live blended scores.
Trend engagement	Converts 30-day trend statistics into plain-language risk direction for public or planning users.

4 Demonstration

4.1 Demo Setup

The live demo runs against the production VTTSI deployment monitoring 18 Falls Church intersections. Attendees interact with the Vite dashboard in a browser; the operations view shows KPIs, network risk, selected-site details, priority queue, and SafetyChat, while analysis panels preserve the chat dock.

If GPT-4o is unavailable, the briefing degrades to deterministic text from latest safety scores. Regression coverage includes 73 local backend tests, 7 cloud backend smoke tests, and 4 Playwright cloud UI tests checking live backend data, map markers, panel reachability, and layout overlap.

5 Related Work

VTTSI builds on EB crash models [7, 9], MCDM safety indices [1, 2], and surrogate safety indicators [4, 11]. TrafficGPT [12], LLMLight [5], and GPT-Driver/DriveVLM [6, 13] address traffic foundation models, signal control, and AV planning, but not conversational explanation of a live PostGIS-backed infrastructure safety index. VTTSI-Chat uniquely combines a validated hybrid geospatial safety index, tool-augmented LLM access to live API data, and causal spatial narratives.

6 Conclusion

VTTSI-Chat is a working agentic GeoAI system for grounded natural language safety narratives over live spatio-temporal intersection data. Its spatially joined crash baseline, 15-minute telemetry, guarded LLM tools, and Vite dashboard provide a deployable foundation for grounded conversational traffic-safety reasoning.

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